

Objective:

To demonstrate Array index out of bound exception

Execution Steps:

STEP 1:

Open IntelliJ IDEA.

STEP 2:

Create a Java project.

STEP 3:

Create a package named:

javacore

STEP 4:

Create a Java class named:

ArrayIndexOutOfBoundsException

STEP 5:

Write the given program in the class.

STEP 6:

Save the program.

STEP 7:

Compile the program.

STEP 8:

Run the program.

STEP 9:

Execution starts from the main() method.

STEP 10:

The array

{10, 20, 30}

is created.

STEP 11:

The valid array indexes are:

0, 1 and 2.

STEP 12:

The try block tries to access:

arr[5]

STEP 13:

Index 5 is invalid, so

ArrayIndexOutOfBoundsException occurs.

STEP 14:

The catch block handles the exception.

STEP 15:

"Array Index Out Of Bounds Exception" is displayed.

STEP 16:

"Program ended" is displayed.

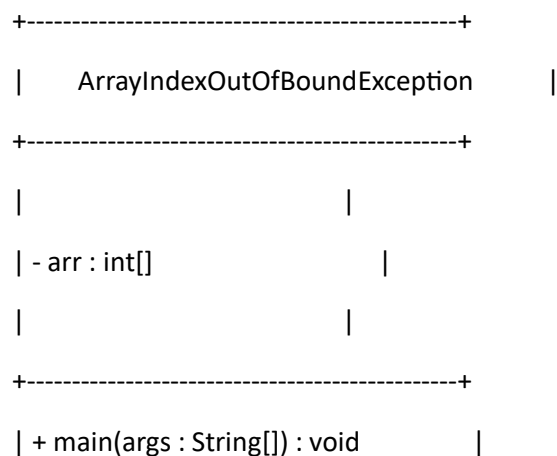
STEP 17:

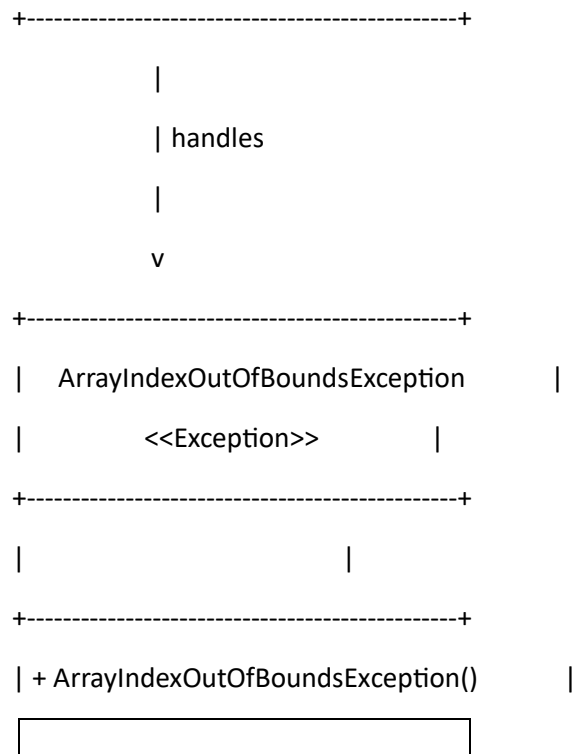
The program ends.

Algorithm:

1. Start the program.
2. Create an integer array with the values 10, 20 and 30.
3. Start the try block.
4. Try to access the element at index 5.
5. Check whether index 5 is valid.
6. Since the valid indexes are only 0, 1 and 2, `ArrayIndexOutOfBoundsException` occurs.
7. Catch the exception using the catch block.
8. Display "Array Index Out Of Bounds Exception".
9. Display "Program ended".
10. Stop the program.

UML Class diagram:





Code:

```
package javacore;
```

```
public class ArrayIndexOutOfBoundsException {
```

```
    public static void main(String[] args) {
```

```
        int arr[] = {10, 20, 30};
```

```
        try {
```

```
            System.out.println(arr[5]);
```

```
        }
```

```
        catch (ArrayIndexOutOfBoundsException e) {
```

```
            System.out.println("Array Index Out Of Bounds Exception");
```

```
        }
```

```
        System.out.println("Program ended");
```

```
    }
```

```
}
```

Output:

Array Index Out Of Bounds Exception

Program ended